



Modernizing Child Welfare Technologies and Tools: Opportunities for Predictive Risk Modeling to Improve Child Safety and Outcomes

Cody Inman – Delegated Commissioner, Administration on Children, Youth, and Families, Administration for Children and Families, HHS

Alex J. Adams – Assistant Secretary, Administration for Children and Families, HHS

Key Points

- Most child welfare agencies in the United States depend on manual tools to assess children’s risk of abuse and neglect. These tools are time consuming to complete, prone to human error and influence, and rely on pre-determined questions and weights that have minimal predictive accuracy.
- Automated predictive risk models represent an important advancement in risk assessment, relying on information already captured in case management systems. They generate real-time risk classifications that are more accurate, more consistent, and far less resource-intensive than traditional manual tools.
- In the last decade, predictive risk models and predictive analytics, more generally, have been successfully implemented to support child welfare staff across the decision-making continuum. Use cases include or could include:
 - Hotline screening of maltreatment allegations;
 - Supervisory oversight of active child abuse and neglect investigations;
 - Monitoring in-home cases to promote safety and well-being;
 - Targeting real-time quality assurance and accountability reviews;
 - Informing risk-based staffing assignments;
 - Increasing caregiver recruitment and retention rates for both relatives and non-relatives; and
 - Improving caregiver and child matching.
- Findings from randomized controlled trials and quasi-experimental studies indicate that these tools have the potential to improve child safety,^{1,2} create time savings for workers,³ reduce disparities,^{1,4} and produce greater visibility and accountability for decisions.
- By integrating predictive risk modeling into child welfare screening and assessment processes, states can better fulfill their responsibility to provide consistent and data-driven services that improve outcomes for children and families.

Background

Child welfare agencies face the difficult task of responding to immediate child safety concerns, while also working to prevent future harm. High maltreatment referral volumes, staffing shortages, and inconsistent workforce expertise creates significant variability in: (1) the information incorporated into maltreatment referral screening decisions; (2) the quality of child abuse and neglect investigations; (3) the degree of monitoring when cases are opened for services; and (4) agency-wide supervisory oversight, quality assurance,

and agency resource-targeting. The pressures faced by child welfare agencies are not new, but they are becoming more acute as child welfare agencies are expected to operate with greater precision during the limited windows in which child safety assessments and early service decisions occur.⁵

Yet although we ask caseworkers to make data-informed decisions, the systems they rely on rarely make it easy or efficient for them to gather the information they need. The primary function of child welfare case-management systems — whether older legacy platforms or newer comprehensive child welfare information system (CCWIS) designs — is to capture and store information. As a result, these systems operate as digital filing cabinets, requiring staff to spend time and countless clicks digging through records to reconstruct family histories that are essential for understanding the nature of current safety concerns and future risk of harm.

One state audit found that a worker had to click through 42 screens just to determine whether an adult had a prior history as a child abuse perpetrator.⁶ Modernizing child welfare technology so it surfaces critical information, rather than simply storing it, is essential to helping caseworkers make timely, accurate decisions.

President Donald J. Trump and First Lady Melania Trump’s Executive Order on Fostering the Future for American Children and Families includes a directive to expand states’ use of technological solutions, including predictive analytics and tools powered to increase caregiver recruitment and retention rates, improve caregiver and child matching, and deploy Federal child-welfare funding to maximally effective purposes and recipients.

Further, in light of the Administration for Children and Families’ (ACF) strategic priority to ensure a safe and stable home for every child, child welfare system tools and technologies should support more precise and proactive decision-making. ACF has emphasized strengthening the balance between the number of available foster homes and the number of children who need placement. States can improve this balance --- or the ratio of foster homes to children --- by strategically increasing the number of foster homes, decreasing the number of children in need of foster care, or both.⁷ Predictive risk modeling, in combination with other predictive analytics tools, can help states make progress on both sides of this equation.

ACF held a stakeholder roundtable on predictive risk modeling in December 2025 to explore key considerations and learn from jurisdictions deploying these predictive risk modeling tools. This issue brief summarizes insights from the roundtable discussion, follow-up conversations, and relevant literature.

Throughout this brief, the term “predictive risk model” (or PRM) refers to a specific class of algorithms that use historical and current data points to generate child-level estimates that a specific outcome (e.g., future foster care entry) will occur. In contrast, “predictive analytics” is a broader term that encompasses a range of analytic methods —including PRM —used to forecast probabilities of outcomes, changes in count-based trends, and other outputs.

How Is Maltreatment Risk Currently Assessed?

Child welfare agencies make high-stakes decisions every day. But the field has been cautious to leverage important data and technology adaptations. Most child welfare agencies use assessment tools that consist of a standard set of structured questions that are completed manually by workers. Workers record answers, which are then scored using pre-set rules that assign simple weights to each response (e.g., 0, 1, 2). The total score is then used to estimate a child’s level of risk and help guide service decisions.⁸ When initially developed in the late 1990s, these tools represented an important step forward at a time when child welfare agencies had been almost entirely reliant on the judgements of individual staff.^{9,10,11,12}

But these structured, checklist tools were created under constraints that no longer apply in today’s data-rich and more technologically advanced environment.^{13,14} Older tools depend on workers manually completing and

scoring a brief checklist of items, failing to incorporate the breadth of information already stored in modern case management systems.¹⁵ Because these tools are completed manually, assessments are typically conducted only once or twice during an investigation or open case, even though child safety conditions and risk dynamics often change. In addition, these tools are prone to human error and manipulation. They contribute to a system in which critical child safety and family service decisions may vary based on subjective judgments embedded in the checklist.¹⁶ Some checklist items require workers to interpret behavior or intent -- for example, deciding whether a parent minimized harm or blamed a child for a maltreatment incident. As a result, children with very different risk profiles may receive similar scores and responses, and critical decisions can hinge on incomplete or inconsistently applied information.¹⁷

For years, audits and other case reviews have highlighted “practice failures” that have compromised the safety of children.^{18,19} More often than not, these include missing, improperly completed, or late risk and safety assessments.²⁰ As one state audit noted: *“the department has unnecessarily risked the health and safety of children in its care because it has not consistently completed investigations and required safety and risk assessments on time or accurately. As a result, the department has left children in unsafe and abusive situations for months longer than necessary.”*²¹ In a 2024 audit by the U.S. Department of Health and Human Services’ Office of the Inspector General, titled *“Maine Did Not Comply With Screening, Assessment, and Investigation Requirements for Responding to Reports of Child Abuse and Neglect”*, auditors concluded that there was a *“material effect on the safety and best interest of the children”*.²²

What Are the Reported Benefits of Predictive Risk Modeling?

Automated predictive risk models (PRM) help address the shortcomings of current decision-support tools by systematically analyzing the full administrative record stored in child welfare case management systems. Information in these records has been entered by caseworkers and includes details from prior reports of alleged maltreatment, investigation findings, service episodes, and foster care placements. Together, they provide a longitudinal history of child maltreatment dynamics for each child and adult. By identifying patterns in historical data, PRMs estimate the likelihood that specific future events will occur.^{23,24}

Because these models pull information directly from existing records, they can incorporate hundreds of data elements and update in real time as safety and other information changes. The ability to draw on more data, coupled with the fact that these models can be trained and validated locally,²⁵ means they can make accurate classifications quickly, without adding burden to frontline staff. In this way, PRMs offer a more reliable and up-to-date foundation for child welfare decision-making, supporting child welfare agencies’ efforts to protect children while allocating resources more effectively. An additional benefit of these automated tools is that the coded features can be presented back to workers as contextual insights. Information that may otherwise have taken multiple clicks, and significant time, through the case management system is produced within seconds.

PRMs offer clear advantages over manual risk assessment tools. Research has documented considerable inconsistency in child protection decision-making across workers assessing similar cases,^{26,27} as sentiment echoed by workers who report difficulty in completing the tools accurately.²⁸ Even in settings where a structured decision-making approach intends to standardize decision making, inter-rater agreement on child welfare case decisions remains low.²⁹

To confirm, “automated” does not mean that the *decisions* informed by a PRM tool are automated. These tools highlight information that may be relevant, but they do not dictate what should happen in a case. Final decisions remain the responsibility of trained professionals, who interpret model outputs alongside their own assessment, supervisory guidance, and agency policy.

How Can Predictive Risk Modeling Be Used in Child Welfare?

There has been longstanding interest in using PRMs to strengthen the decisions made by the child welfare workforce. These tools are designed to reduce staff burden by automatically assembling and analyzing the information already stored in case management systems, giving workers and supervisors a clearer picture of risk and case complexity. By surfacing complex and high-risk cases quickly and consistently, PRMs help direct attention to the investigations where delays or gaps in information-gathering are most likely to jeopardize child safety.

Importantly, the field has evolved from focusing solely on identifying children at highest risk of harm to also recognizing the value of identifying low-risk families who may not require an investigation or ongoing involvement. This dual capacity — spotting both high-risk and low-risk situations — allows child welfare agencies to reduce unnecessary system contact, create more consistent decision-making, and ensure that staff efforts are concentrated where they will have the greatest impact. These benefits are reflected in how predictive analytics are used across key points in the workflow.

Staff Applications

- **Child Protection Hotline Screening Decisions.** PRM tools help staff make more consistent and accurate decisions about whether to investigate a maltreatment allegation, reducing unnecessary responses to low-risk referrals while minimizing missed high-risk cases.
- **Maltreatment Investigations.** PRM tools provide staff with real-time alerts that flag investigations requiring closer scrutiny, ensuring that elevated-risk cases receive timely oversight and support.
- **Cases with In-Home Services or Foster Care.** More broadly, predictive analytic tools supply staff with up-to-date information on case complexity and risk, track ongoing safety for children receiving in-home or foster care services, and signal when new referrals or prolonged case duration require additional attention.

Other Applications

Predictive analytic tools can also provide child welfare agency leadership with consistent, system-wide information on child safety and risk, strengthening oversight and supporting transparent, data-driven accountability across regions, units, and staff.

- **Real-Time Quality Assurance.** Predictive analytic tools enable quality assurance teams to select and review specific high-risk cases in real time, rather than only after a critical error or oversight has occurred. This proactive visibility allows child welfare agencies to intervene earlier, strengthen practice consistency, and address emerging concerns before they escalate.
- **Workload Assignment Monitoring.** Predictive analytic tools track complexity-adjusted caseloads at the worker level, supporting more balanced assignments and helping child welfare agencies reduce burnout, improve staff retention, and ensure that high-need cases receive the attention they require.
- **Family First Candidacy.** Predictive analytic tools help identify children who meet federal criteria for candidacy under the Family First Prevention Services Act, providing supervisors with clear, data-informed justification for service pathways and documentation.
- **Caregiver Recruitment, Retention, and Placement Stability.** Predictive analytic tools may help identify prospective caregivers, identify current caregivers in need of additional support, and improve the match between the needs of individual children and the demonstrable capability of the caregivers.

Data, Development, and Technology Considerations

PRM tools can be introduced without large-scale technology investments, but it does demand an understanding of the underlying data on which the models are built. Strong data foundations allow for precise model calibration and careful monitoring of feature drift, with the goal of dependable performance. Importantly, these tools can be and have been implemented within existing agency IT infrastructures.

- **Model Build.** Developing a PRM gives jurisdictions the opportunity to train, test, externally validate, and refine a model before it is used in practice. Because these models are trained using a state's historical administrative data, child welfare agencies can evaluate how well a model performs — overall and for specific subpopulations — without affecting any real-time decisions.
- **Data Sources.** Although integrating data across systems (e.g., education, behavioral health, criminal justice) can enhance the information available to child welfare workers, such integration is not required to develop accurate PRM tools. Evidence to date indicates that models built using only child welfare records can achieve strong predictive accuracy. Consistent with this, most implementations thus far have relied exclusively on child welfare administrative data.
- **Data Security.** Ensuring the confidentiality of client records is always a central concern in child welfare, and PRM tools can be designed in ways that minimize risk and comply with applicable statutory and regulatory requirements. Unlike some commercial software-as-a-service systems (SaaS), all predictive analytics models in child welfare can be built, validated, and deployed in the state's own IT environment. This approach addresses a common concern about predictive analytic tools in child welfare by ensuring that confidential and personally identifiable information is always under state control.
- **Case Management Transitions.** Because risk models operate on a separate analytic layer, these tools can be built, tested, and deployed even while a state is upgrading or replacing its child welfare case management system. Several jurisdictions have continued to use predictive tools seamlessly during multi-year CCWIS modernization efforts.

Critiques of Predictive Risk Modeling

Concerns about the use of predictive analytics in child welfare often focus on the data used to develop these models. Critics point out that child welfare data reflect historical decisions that are biased.³⁰ If certain communities have been investigated more frequently or more intensively, a model trained solely on those data may “learn” patterns that reflect system behavior rather than patterns of actual harm.³¹ For this reason, many experts emphasize the importance of validating predictive risk models against outcomes that are not solely derived from child welfare processes, such as hospitalizations for injuries and other objective indicators of serious harm.^{32,33}

Many of these same critiques apply to the traditional, checklist-based risk assessment tools currently used in most states. These tools also rely on administrative data shaped by historical practices and include items such as past allegations of abuse and neglect parental histories of domestic violence, or certain household characteristics. Moreover, because traditional tools rely on workers manually scoring a limited set of pre-defined checklist items — established by tool developers or agencies — each item accounts for a sizable portion of the overall score. When one response carries that level of influence, variation in interpretation or scoring can meaningfully alter the final risk assessment.³⁴ In this sense, the debate must focus not on whether predictive analytics introduce bias, but on how openly those biases are examined and whether the tools in use—old or new—are held to clear standards for transparency, accuracy, and consistency.³⁵

A second category of critique focuses on the role of predictive analytic tools in frontline practice, as a claimed form of “surveillance” and “poverty profiling”. Yet child welfare agencies are required to screen and make

decisions about referrals of alleged maltreatment; PRM tools are designed to aid frontline staff in their required assessments.^{36,37} Critics also raise concerns that if caseworkers rely too heavily on automated scoring systems, their professional judgment could be weakened or key contextual information may be overlooked.³⁸ Predictive analytic tools are not meant to replace the nuanced decision-making that experienced staff bring to their work, but rather to support it.³⁹ Ensuring that any tool — whether manual or predictive — is integrated thoughtfully into practice is essential to maintaining both accuracy and consistency in child welfare decisions.

The Allegheny County Department of Health and Human Services in Pennsylvania is an early adopter of PRM methods. Their development and implementation of its tools was guided by careful consideration of common critiques.^{40,41,42} Allegheny mitigated risks through community engagement, staff training, careful consideration of use cases, and continued monitoring and ongoing quality assurance. Future PRM tool development should consider these issues. When implemented thoughtfully and transparently, the gains in increased accuracy for identifying serious safety concerns outweigh the potential risks.

Lessons Learned From Predictive Risk Modeling in Practice

Although PRM is not yet widely used in child welfare, experience from the field has generated reliable lessons about how these tools can be incorporated responsibly and effectively. The accumulated evidence shows that these tools can enhance decision-making. The field is at a point where informed, responsible expansion is both possible and warranted. The following lessons learned point to a maturing field in which the use of various predictive analytic methods is neither a one-off novelty nor a standalone solution in child welfare.

- Predictive analytics is not a substitute for a strong child welfare workforce trained to gather and evaluate information needed to guide decisions.⁴³ The use of these tools must be embedded in well-defined child welfare agency processes—such as enhanced consultation or differential oversight pathways—to translate risk information into meaningful practice changes. Including corresponding protocols can increase analytic outputs impact and mitigate misinterpretations.
- Stakeholders and communities expect visibility into how models are developed, what information they use, and how results will — or will not — affect decisions. Many jurisdictions that have successfully implemented predictive analytic tools have been intentional in their release of public documentation and community engagement as core components of model governance.⁴⁴
- Routine recalibration, review of performance by subpopulation, and structured feedback loops help ensure the tool remains accurate, fair, and aligned with agency goals.⁴⁵

Conclusions

Predictive risk modeling represents an important modernization of long-standing risk assessment practices. Because these tools are automated, locally validated, and built on the same information workers rely on every day, they can generate real-time insights that support consistent, efficient, and timely decision-making. When paired with strong practice protocols, workforce training, and transparent governance, these tools provide a clearer, more comprehensive view of risk and case complexity—precisely when that information is most needed. States that adopt and thoughtfully implement these tools will be better positioned to protect children, serve families, and deliver a more responsive and effective child welfare system.

Appendix: How Are Child Welfare Agencies Currently Using Predictive Risk Modeling?

Several jurisdictions have identified predictive risk modeling as a key modernization strategy, enabling them to direct staff and resources toward the children and families who need them most while appropriately diverting lower-risk families to community supports. Examples of currently implemented tools are presented below.

Allegheny County, Pennsylvania

In 2016, the Allegheny Family Screening Tool (AFST) became the first PRM implemented by a public child welfare agency. The AFST uses historical child welfare and cross-system data (e.g., criminal justice, behavioral health) to support child protection hotline screening decisions made jointly by a call screener and their supervisor. Prior to the implementation of the AFST, staff had to manually search for relevant historical data for clients, with little guidance for how that information would inform the agency's decision to investigate. Analyses indicated that they were screening out more than a quarter of their highest risk referrals - with many of those children later reported having experienced more significant harm. They also screened-in nearly 50 percent of low-risk referrals— a poor use of child welfare resources, often creating unnecessary interactions with families.

Model development and implementation included strong public-facing documentation, community engagement, and third-party evaluations. Multiple independent evaluations of the AFST have produced important evidence about how predictive analytics can strengthen screening decisions. Studies have shown that the tool is well-calibrated, meaning that higher scores are strongly associated with increased likelihood of subsequent serious harm including maltreatment-related hospitalizations, while also significantly reducing disparities in decision-making.

Arapahoe, Douglas, and Larimer Counties, Colorado

In Colorado, Arapahoe, Douglas, and Larimer counties are also experimenting with predictive analytics for decision support. The counties' predictive risk model uses data from the State's CCWIS/SACWIS system and includes a wide range of features related to child-, parent-, and person responsible for abuse and neglect— carefully weighted and iteratively refined. These data generate risk estimates that inform decision-making and supervisory review at all stages, including at call screening, assessment, and the oversight of open in-home and foster care cases. Supervisors receive a consolidated summary of case-management system information, paired with measures of risk and case complexity, enhancing their ability to monitor workloads and prioritize supervisory support. Ultimately, the model supports clinical supervision but does not replace human judgement.

Independent randomized controlled trials were conducted in Douglas and Larimer Counties to assess the impact of using these risk models to support the team-based evaluation of referrals. Findings suggest that the use of the Colorado predictive risk model reduced time spent making decisions,⁴⁶ improved the safety of children, and reduced disparities in outcomes.⁴⁷

Idaho (Statewide)

Following its 2020 implementation of the CCWIS, Idaho modernized its centralized intake and investigation workflow by deploying an automated predictive risk model. The model provides supervisors with decision support during initial child protection hotline screening by highlighting referrals with the highest predicted risk and identifying those with the lowest predicted risk. Referrals classified as low risk may be appropriate for screened-out or diverted to alternative service pathways, allowing staff to focus resources on cases with greater safety concerns.

Once a referral is accepted for investigation, Idaho continues to use the predictive risk model to support statewide supervisory oversight. The tool generates consistent, data-informed risk and safety intelligence and supports workload management by helping supervisors identify cases that require more intensive investigation or casework. This approach is designed to increase consistency and help ensure that investigative resources align with the level of predicted risk.

Los Angeles County, California

In 2021, the Los Angeles County Department of Children and Family Services launched a predictive risk modeling pilot in three regional offices to support supervisory oversight of child maltreatment investigations. The model draws on historical and current information from the state's child welfare case management system to identify a small subset of investigations with elevated likelihood of child harm leading to future system involvement. Rather than being used to screen child protection hotline calls or dictate service decisions, the tool is designed to give supervisors earlier, more consistent access to relevant case history to support engagement, service coordination, and management of investigations with very different risk and service needs.

Model development and implementation were co-designed with supervisors in the pilot offices and accompanied by public documentation, community engagement, and external review. Analyses of pilot data showed improvements in measures of child safety, including fatality and near-fatality outcomes, increased consultations and casework activities during the investigation, and reductions in re-reports among children who remained at home in pilot offices compared with non-pilot offices. In pilot offices, there was no increase in disparities or foster care placements following adoption of the tool. Supervisors reported that the model helped prioritize case consultations and improve communication across units. In parallel, the county is also exploring use of the model to identify investigations that were screened in at the child protection hotline but with closer review and additional information-gathering, may be safely evaluated out.

References

- ¹ M.-P. Grimon and C. Mills, “Better Together? A Field Experiment on Human–Algorithm Interaction in Child Protection,” *arXiv* (2025), <https://doi.org/10.48550/arXiv.2502.08501>
- ² Los Angeles County Department of Children and Family Services, “New Analysis of DCFS Data Shows Improvement in Safety with Use of Data-Informed Technology,” September 5, 2024, <https://dcfs.lacounty.gov/press-release-new-analysis-of-dcfs-data-shows-improvement-in-safety-with-use-of-data-informed-technology/>
- ³ M. D. Fitzpatrick, K. Sadowski, and C. Wildeman, “Algorithms and Decision-Making: Evidence from Child Maltreatment Reports,” *Journal of Human Resources* (2025), <https://jhr.uwpress.org/content/early/2025/08/01/jhr.0224-13437R2>.
- ⁴ K. Rittenhouse, E. Putnam-Hornstein, and R. Vaithianathan, “Algorithms, Humans and Racial Disparities in Child Protection Systems: Evidence from the Allegheny Family Screening Tool” (2024), <https://krittenh.github.io/katherine-rittenhouse.com/Algorithms%20Humans%20Racial%20Disparities.pdf>
- ⁵ K. Rittenhouse, D. Simon, and L. Lacey, “Child Maltreatment Investigations and Family Well-Being,” NBER Working Paper No. 34334 (Cambridge, MA: National Bureau of Economic Research, 2025), <https://www.nber.org/papers/w34334>
- ⁶ Oregon Secretary of State, Audits Division, *Oregon’s Child Welfare Information System Is Adequate for Case Management, but Enhancements Are Needed to Improve Usability*, Report No. 2020-01 (2020), <https://records.sos.state.or.us/ORSOSWebDrawer/Recordhtml/7132832>
- ⁷ A. J. Adams, “Delivering on President Trump’s Call to Strengthen Foster Care: A Home for Every Child,” *The Daily Signal*, November 21, 2025, <https://www.dailysignal.com/2025/11/21/delivering-on-president-trumps-call-to-strengthen-foster-care-a-home-for-every-child/>
- ⁸ S. Cuccaro-Alamin, R. Foust, R. Vaithianathan, and E. Putnam-Hornstein, “Risk Assessment and Decision Making in Child Protective Services: Predictive Risk Modeling in Context,” *Children and Youth Services Review* 79 (2017): 291–298, <https://doi.org/10.1016/j.childyouth.2017.06.027>
- ⁹ H. J. Doueck, M. Levine, and D. E. Bronson, “Risk Assessment in Child Protective Services: An Evaluation of the Child at Risk Field System,” *Journal of Interpersonal Violence* 8, no. 4 (1993): 446–467, <https://doi.org/10.1177/088626093008004002>
- ¹⁰ C. R. McNellan, D. J. Gibbs, A. S. Knobel, and E. Putnam-Hornstein, “The Evidence Base for Risk Assessment Tools Used in U.S. Child Protection Investigations: A Systematic Scoping Review,” *Child Abuse & Neglect* 134 (2022): 105887, <https://doi.org/10.1016/j.chiabu.2022.105887>
- ¹¹ M. A. Johnson, S. Stone, C. Lou, C. M. Vu, J. Ling, P. Mizrahi, and M. J. Austin, “Family Assessment in Child Welfare Services,” *Journal of Evidence-Based Social Work* 5, no. 1–2 (2008): 57–90, <https://doi.org/10.1080/15433710802301158>
- ¹² C. Baird, D. Wagner, T. Healy, and K. Johnson, “Risk Assessment in Child Protective Services: Consensus and Actuarial Model Reliability,” *Child Welfare* 78, no. 6 (1999): 723–748.
- ¹³ R. Vaithianathan, S. Cuccaro-Alamin, and E. Putnam-Hornstein, “Improving Child Welfare Practice Through Predictive Risk Modeling: Lessons from the Field,” in *Strengthening Child Safety and Well-Being Through Integrated Data Solutions*, eds. C. M. Connell and D. M. Crowley (Cham: Springer, 2023), 115–126, https://doi.org/10.1007/978-3-031-36608-6_8
- ¹⁴ M. J. Camasso and R. Jagannathan, “Modeling the Reliability and Predictive Validity of Risk Assessment in Child Protective Services,” *Children and Youth Services Review* 22, nos. 11–12 (2000): 873–896, [https://doi.org/10.1016/S0190-7409\(00\)00121-3](https://doi.org/10.1016/S0190-7409(00)00121-3)
- ¹⁵ R. Vaithianathan, S. Cuccaro-Alamin, and E. Putnam-Hornstein, “Improving Child Welfare Practice Through Predictive Risk Modeling: Lessons from the Field,” in *Strengthening Child Safety and Well-Being Through Integrated Data Solutions*, eds. C. M. Connell and D. M. Crowley (Cham: Springer, 2023), 115–126, https://doi.org/10.1007/978-3-031-36608-6_8
- ¹⁶ North Carolina General Assembly, Program Evaluation Division, *Child Protective Services Intake Screening Lacks Consistency*, Report No. 2019-10 (2019), <https://webservices.ncleg.gov/ViewDocSiteFile/21800>
- ¹⁷ State Auditor of Missouri, *Audit of Child Abuse and Neglect Reporting and Response System*, Report No. 2000-132 (2000), <https://auditor.mo.gov/press/2000-132.pdf>

- ¹⁸ Office of Inspector General, U.S. Department of Health and Human Services, *West Virginia Did Not Comply with Intake, Screening, Assessment, and Investigation Requirements for Responding to Reports of Child Abuse and Neglect*, Audit Report No. OAS-25-01-011 (2025), <https://oig.hhs.gov/documents/audit/11268/OAS-25-01-011.pdf>
- ¹⁹ Arizona Office of the Auditor General, *Arizona Department of Child Safety—Child Safety, Removal, and Risk Assessment Practices*, Special Report No. 15-118 (2015), https://www.azauditor.gov/sites/default/files/2023-11/15-118_Report.pdf
- ²⁰ Illinois Office of the Auditor General, *State Compliance Examination: Illinois Department of Children and Family Services, for the Two Years Ended June 30, 2024* (2025), <https://www.auditor.illinois.gov/Audit-Reports/Compliance-Agency-List/DCFS/FY24-DCFS-Comp-Full.pdf>.
- ²¹ California State Auditor, *Los Angeles County Department of Children and Family Services: It Has Not Adequately Ensured the Health and Safety of All Children in Its Care*, Report No. 2018-126 (2019), <https://information.auditor.ca.gov/pdfs/reports/2018-126.pdf>
- ²² U.S. Department of Health and Human Services, Office of Inspector General, *Maine Did Not Comply with Screening, Assessment, and Investigation Requirements for Responding to Reports of Child Abuse and Neglect*, Report No. A-01-23-02500 (2024), <https://oig.hhs.gov/documents/audit/10044/A-01-23-02500.pdf>
- ²³ R. Vaithianathan, T. Maloney, E. Putnam-Hornstein, and N. Jiang, “Children in the Public Benefit System at Risk of Maltreatment: Identification via Predictive Modeling,” *American Journal of Preventive Medicine* 45, no. 3 (2013): 354–359, <https://doi.org/10.1016/j.amepre.2013.05.011>
- ²⁴ B. Drake, M. Jonson-Reid, M. G. Ocampo, M. Morrison, and D. Dvalishvili, “A Practical Framework for Considering the Use of Predictive Risk Modeling in Child Welfare,” *The ANNALS of the American Academy of Political and Social Science* 692, no. 1 (2020): 162–181, <https://doi.org/10.1177/0002716220978200>
- ²⁵ R. Vaithianathan, E. Putnam-Hornstein, A. Chouldechova, D. Benavides-Prado, and R. Berger. “Hospital Injury Encounters of Children Identified by a Predictive Risk Model for Screening Child Maltreatment Referrals: Evidence from the Allegheny Family Screening Tool.” *JAMA Pediatrics* 174, no. 11 (2020), e202770, <https://doi.org/10.1001/jamapediatrics.2020.2770>
- ²⁶ P. H. Rossi, J. Schuerman, and S. Budde, “Understanding Decisions about Child Maltreatment,” *Evaluation Review* 23, no. 6 (1999): 579–598, <https://doi.org/10.1177/0193841X9902300601>
- ²⁷ E. Munro, “Common Errors of Reasoning in Child Protection Work,” *Child Abuse & Neglect* 23, no. 8 (1999): 745–758, [https://doi.org/10.1016/S0145-2134\(99\)00053-8](https://doi.org/10.1016/S0145-2134(99)00053-8)
- ²⁸ E. A. Bosk, “What Counts? Quantification, Worker Judgment, and Divergence in Child Welfare Decision Making,” *Human Service Organizations: Management, Leadership & Governance* 42, no. 2 (2018): 205–224, <https://doi.org/10.1080/23303131.2018.1446588>
- ²⁹ C. Bartelink, T. A. van Yperen, and I. J. ten Berge, “Deciding on Child Maltreatment: A Literature Review on Methods That Improve Decision-Making,” *Child Abuse & Neglect* 49 (2015): 142–153, <https://doi.org/10.1016/j.chiabu.2015.07.002>
- ³⁰ C. S. Greeley, “Child Maltreatment Prevention in the Era of Coronavirus Disease 2019,” *JAMA Pediatrics* 174, no. 11 (2020): e202776, <https://doi.org/10.1001/jamapediatrics.2020.2776>
- ³¹ A. Samant, “Automated Decision Systems: Child Welfare Predictive Analytics Tools,” (American Civil Liberties Union of Washington: July 11, 2023, <https://www.aclu-wa.org/news/automated-decision-systems-child-welfare-predictive-analytics-tools/>
- ³² R. Vaithianathan, E. Putnam-Hornstein, A. Chouldechova, D. Benavides-Prado, and R. Berger, “Hospital Injury Encounters of Children Identified by a Predictive Risk Model for Screening Child Maltreatment Referrals: Evidence from the Allegheny Family Screening Tool,” *JAMA Pediatrics* 174, no. 11 (2020): e202770, <https://doi.org/10.1001/jamapediatrics.2020.2770>
- ³³ J. Kelly, “Yonatan’s Death Spurs Deep Look at How L.A. Gauges Child Abuse Risk,” *The Imprint*, November 1, 2016, <https://imprintnews.org/featured/yonatans-death-spurs-deep-look-l-gauges-child-abuse-risk/22452>
- ³⁴ B. Drake, M. Jonson-Reid, M. G. Ocampo, M. Morrison, and D. Dvalishvili, “A Practical Framework for Considering the Use of Predictive Risk Modeling in Child Welfare,” *The ANNALS of the American Academy of Political and Social Science* 692, no. 1 (2020): 162–181, <https://doi.org/10.1177/0002716220978200>
- ³⁵ K. Rittenhouse, E. Putnam-Hornstein, and R. Vaithianathan, “Algorithms, Humans and Racial Disparities in Child Protection Systems: Evidence from the Allegheny Family Screening Tool” (2024), <https://krittenh.github.io/katherine-rittenhouse.com/Algorithms%20Humans%20Racial%20Disparities.pdf>

- ³⁶ A. Samant, A. Horowitz, K. Xu, and S. Beiers, *Family Surveillance by Algorithm: The Rapidly Spreading Tools Few Have Heard Of* (American Civil Liberties Union, September 29, 2021), <https://www.aclu.org/documents/family-surveillance-algorithm>
- ³⁷ V. Eubanks, “A Child Abuse Prediction Model Fails Poor Families,” *WIRED*, January 15, 2018, <https://www.wired.com/story/excerpt-from-automating-inequality/>
- ³⁸ M. Gerchick, T. Jury, T. Shah, A. Gutierrez, S. Beiers, N. Shemtov, K. Xu, A. Samant, and A. Horowitz, *The Devil Is in the Details: Interrogating Values Embedded in the Allegheny Family Screening Tool* (American Civil Liberties Union, 2023), <https://www.aclu.org/the-devil-is-in-the-details-interrogating-values-embedded-in-the-allegheny-family-screening-tool>
- ³⁹ V. Eubanks, “A Child Abuse Prediction Model Fails Poor Families,” *WIRED*, January 15, 2018, <https://www.wired.com/story/excerpt-from-automating-inequality/>
- ⁴⁰ T. Dare and E. Gambrill, *Ethical Analysis: Predictive Risk Models at Call Screening for Allegheny County* (2017), <https://www.alleghenycountyanalytics.us/wp-content/uploads/2019/05/Ethical-Analysis-16-ACDHS-26-PredictiveRisk-Package-050119-FINAL-2.pdf>.
- ⁴¹ Allegheny County Department of Human Services, *Ethical Analysis: Predictive Risk Models at Call Screening for Allegheny County: Response by the Allegheny County Department of Human Services* (2017), <https://www.alleghenycountyanalytics.us/wp-content/uploads/2019/05/Response-to-Ethical-Analysis-from-16-ACDHS-26-PredictiveRisk-Package-050119-FINAL-3.pdf>
- ⁴² J. D. Goldhaber-Fiebert and L. Prince, *Impact Evaluation of a Predictive Risk Modeling Tool for Allegheny County’s Child Welfare Office* (2019), <https://www.alleghenycountyanalytics.us/wp-content/uploads/2019/05/Impact-Evaluation-from-16-ACDHS-26-PredictiveRisk-Package-050119-FINAL-6.pdf>
- ⁴³ M.-P. Grimon and C. Mills, “Better Together? A Field Experiment on Human–Algorithm Interaction in Child Protection,” *arXiv* (2025), <https://doi.org/10.48550/arXiv.2502.08501>
- ⁴⁴ Y. Roberts, K. O’Brien, and P. Pecora, *Considerations for Implementing Predictive Analytics in Child Welfare* (Casey Family Programs, April 2018), <https://www.casey.org/media/Considerations-for-Applying-Predictive-Analytics-in-Child-Welfare.pdf>
- ⁴⁵ D. Whicher, E. Pendl-Robinson, K. Jones, and A. Kalisher, *Avoiding Racial Bias in Child Welfare Agencies’ Use of Predictive Risk Modeling* (Office of Human Services Policy, U.S. Department of Health and Human Services, September 29, 2022), <https://aspe.hhs.gov/reports/child-welfare-predictive-risk-models>
- ⁴⁶ M. D. Fitzpatrick, K. Sadowski, and C. Wildeman, “Algorithms and Decision-Making: Evidence from Child Maltreatment Reports,” *Journal of Human Resources*, advance online publication (2025), <https://jhr.uwpress.org/content/early/2025/08/01/jhr.0224-13437R2>
- ⁴⁷ M.-P. Grimon and C. Mills, “Better Together? A Field Experiment on Human–Algorithm Interaction in Child Protection,” *arXiv* (2025), <https://doi.org/10.48550/arXiv.2502.08501>

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Administration for Children & Families

330 C Street, S.W.
Washington, D.C. 20201

SUGGESTED CITATION

Inman, C., and Adams, AJ. Modernizing Child Welfare Technologies and Tools: Opportunities for Predictive Risk Modeling to Improve Child Safety and Outcomes. Administration for Children and Families, U.S. Department of Health and Human Services. March 2026.

COPYRIGHT INFORMATION

All material appearing in this report is in the public domain and may be reproduced or copied without permission; citation as to source, however, is appreciated.

DISCLOSURE

This communication was printed, published, or produced and disseminated at U.S. taxpayer expense.

Links and references to information from non-governmental organizations are provided for informational purposes and are not an HHS endorsement, recommendation, or preference for the non-governmental organizations.